

Problem statement

Problem Statement No.1: Neocare Agent

The Challenge

Newborns require continuous monitoring and specialized care during the first 28 days of life, yet many parents lack the knowledge and tools to monitor feeding, sleep, temperature, growth, and warning signs of illness. Delayed identification of health issues and inadequate access to healthcare guidance contribute to preventable neonatal complications. There is a need for an affordable, user-friendly solution that assists caregivers in monitoring newborn health, provides timely alerts, and connects them with healthcare resources.

The Objective

Build an **AI-powered NeoCare Agent** that delivers personalized newborn care guidance and health insights for parents, caregivers, and healthcare professionals. The solution should:

- Monitor and analyze newborn health data such as feeding, sleep, body temperature, diaper changes, and growth.
- Provide personalized recommendations based on the newborn's age, weight, and health status.
- Detect potential health concerns early and generate timely alerts for caregivers.
- Track vaccination schedules, medications, and routine health check-ups with automated reminders.
- Offer evidence-based guidance on breastfeeding, nutrition, hygiene, safe sleep, and newborn care practices.
- Maintain secure digital health records for continuous monitoring and easy sharing with healthcare professionals.
- Support parents with 24/7 AI-powered assistance by answering common newborn care questions and providing actionable recommendations.
- Improve neonatal health outcomes through early intervention, continuous monitoring, and informed decision-making.

Specific Objectives

1. Monitor essential newborn health parameters such as body temperature, heart rate, oxygen saturation (SpO₂), and sleep patterns in real time.
2. Provide instant alerts to parents and healthcare providers when abnormal health conditions are detected.
3. Track feeding schedules, diaper changes, sleep duration, and growth milestones to support healthy development.
4. Send reminders for vaccinations, medications, and regular health check-ups.
5. Offer educational guidance on newborn care, breastfeeding, hygiene, and recognizing danger signs.

6. Store and maintain health records for easy access and monitoring by caregivers and medical professionals.
7. Improve early detection of neonatal health complications, reducing the risk of severe illness and hospitalization.
8. Develop an affordable, reliable, and easy-to-use solution suitable for both hospitals and home environments.Tech Stack (Mandatory)
9. IBM Langflow / IBM Orchestrate+IBM Models